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Memory delay and haptic feedback influence the dissociation of tactile cues for perception and action

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ABSTRACT

The somatosensory processing model (SPM) asserts that dissociable cortical processing streams mediate tactile perceptions and actions via relative and absolute cues, respectively (Dijkerman and de Haan, 2007). Accordingly, we sought to determine whether the introduction of a memory delay and/or physically touching a target object (i.e., haptic feedback) differentially influences the cues supporting tactile perceptions and actions. Participants used their right hand to manually estimate (i.e., perceptual task) or grasp (i.e., action task) differently sized objects placed on the palm of their left limb in conditions wherein the target object was available for the duration of the response (i.e., closed-loop condition), or was removed prior to response cuing (i.e., memory-guided condition). As well, trials were performed in conditions wherein the physical object was available (i.e., haptic feedback) or unavailable (i.e., no haptic feedback) to touch. Notably, we computed just-noticeable-difference (JND) scores to determine whether the aforementioned tasks and conditions adhered to – or violated - the relative properties of Weber's law. JNDs for manual estimations adhered to Weber's law across each condition – a finding supporting the SPM's contention that an immutable and relative percept supports tactile perceptions. In turn, JNDs for grasping violated Weber's law only when haptic feedback was available. Such a finding indicates that haptic feedback supports the absolute calibration between a tactile defined object and the required motor output. What is more, our study highlights that multiple somatosensory cues (i.e., tactile and haptic) support goal-directed grasping.

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1. Introduction

Vision predominates many of the goal-directed activities that we perform on a day-to-day basis. It is, however, important to recognize that tactile cues play a pivotal role in object identification and movement control. For example, a coin placed on the palm of the left hand mechanically deforms the skin and leads to mechanoreceptor-derived (i.e., tactile) feedback supporting *perceptual* identification of the object (Lederman and Klatzky, 2009). Moreover, in executing a *motor* response tactile feedback provides the information necessary to allow the right limb to effectively reach-to-grasp the coin resting in the left palm. Interestingly, Dijkerman and de Haan's (2007) somatosensory processing model (SPM) asserts that tactile perceptions and actions are mediated via functionally and anatomically distinct cortical pathways. In particular, the SPM contends that a ventral pathway extending from the anterior parietal cortex (APC) and the secondary somatosensory cortex (SII) to the posterior insula mediates perceptions. In turn, a dorsal pathway originating from the APC and the SII and extending to the posterior parietal cortex (PPC) is proposed to support goal-directed actions. Moreover, the SPM asserts that the ventral pathway mediates perceptions via *relative* cues wherein the properties of an object (e.g., size) are compared to other objects and/or by accessing memory-based information (i.e., allocentric frame of reference). Thus, a coin on the palm of one's hand can be identified via: (1) concurrent tactile feedback, and/or (2) temporally persistent knowledge from previous tactile experiences. In contrast, the spatial and temporal demands associated with goal-directed actions require that the dorsal stream regulate motor output via *absolute* information related to the size and position of the to-be-grasped object (i.e., egocentric frame of reference) (for review of tactile frames of reference see Lederman and Klatzky, 2009.

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Initial support for the SPM was derived from the clinical neuropsychology literature and the report of a double dissociation for tactile perceptions and actions in persons recovering from unilateral stroke (Anema et al., 2009; see also Paillard et al., 1983). Moreover, recent work by our group (Davarpanah Jazi and Heath, 2014) demonstrated a perception/action dissociation in neurologically intact individuals. In that study, participants used their right hand to manually estimate (i.e., perceptual task) or grasp (i.e., motor task) differently sized objects (20, 30, 40 and 50 mm) placed on the forearm or palm of their left limb, with all responses being completed in the absence of vision[1]. Importantly, just noticeable difference scores (JNDs) were computed to examine the extent to which the different tasks adhered to, or violated, the psychophysical principles of Weber's law. Indeed, Weber's law asserts that the ability to discriminate between an original (e.g., the target stimulus) and a comparator (e.g., grip aperture size) stimulus is proportional to the magnitude of the original stimulus and that the sensitivity of detecting a change in any physical continuum is *relative* as opposed to *absolute*. Thus, a linear increase in JNDs with increasing object size indicates adherence to Weber's law and therefore provides a rule-based demonstration of the processing of relative – but not absolute – object properties (i.e., size). Results for the manual estimation task showed that JNDs increased linearly with increasing object size, whereas JNDs for the grasping task were refractory to object size. In other words, tactile perceptions but not actions adhered to Weber's law – a finding supporting the theoretical tenets of the SPM.

The first goal of the present investigation was to determine the temporal properties by which absolute tactile cues are available to support grasping. The basis for this question stems from Goodale and Milner's (1992) influential duplex model of visual processing (i.e., the perception/action model: PAM). In particular, the PAM asserts that absolute visual information mediated by dedicated visuomotor networks residing in the PPC (i.e., the dorsal *visual* pathway) support goal-directed actions. Moreover, the PAM contends that absolute visual information is available to the motor system only on a moment-to-moment basis (for review see Goodale, 2011). In support of this view, Goodale et al. (1994) reported that an individual with a visual form agnosia (DF) was able to scale her grip aperture to a target object when it was visible at the time of response cuing; however, that ability was lost when the target object was occluded 2000 ms prior to response cuing.[2] As well, extensive work has shown that reaching and grasping efficiency and effectiveness in neurologically intact individuals is diminished following a period of visual delay (e.g., Bridgeman et al., 1997; Elliott and Madalena, 1987; Elliott et al., 1999; Heath, 2005; Heath et al., 2004; Hu et al., 1999; Westwood et al., 2000; for review see Heath et al., 2010). For example, Westwood and Goodale (2003) found that grasping responses were refractory to the relative properties of a pictorial illusion "…*only* after the response is cued, and *only* if the target is visible" (p. 243) (i.e., real-time control hypothesis). According to the PAM, the absence of real-time control results in actions mediated via a temporally stable and relative target percept maintained by visuoperceptual networks in the inferotemporal cortex (i.e., the ventral *visual* pathway).

Based on the findings from the visual domain, we modified the tactile manual estimation and grasping tasks used in our group's previous work (Davarpanah Jazi and Heath, 2014) to include a memory delay. Participants used their right hand to manually estimate or grasp differently sized target objects placed on the palm of their left hand. Importantly, responses were completed in: (1) a closed-loop condition wherein the target object remained on the left palm throughout the response (see also Davarpanah Jazi and Heath, 2014) and, (2) a memory-guided condition wherein the target object was removed from the left palm 2000 ms prior to response cuing and was unavailable throughout the response. Thus, real-time tactile feedback was unavailable in the memory-guided condition. In terms of research outcomes, if manual estimations are supported via a relative – and temporally stable – tactile percept then JNDs in closed-loop and memory-guided conditions should adhere to Weber's law. In other words, it is predicted that a memory-delay will not influence the nature of the tactile cues supporting perceptions. In terms of the grasping tasks, if absolute tactile cues are available to the motor system only on a moment-to-moment basis then JNDs for the closed-loop and memory-guided conditions should respectively violate and adhere to Weber's law. In particular, it is predicted that the absence of real-time tactile feedback will render the processing of object size via the same relative percept as supporting manual estimations.

The second objective of our study was to determine whether *terminal haptic feedback* derived from physically grasping (i.e., touching) an object influences the nature of the tactile information supporting grasping. In this context we emphasize that haptic feedback is different from tactile feedback because the former stems from physically grasping a target object. Notably, haptic feedback is derived from proprioceptive information related to finger and thumb orientation and therefore provides absolute cues related to object size. In the context of the present investigation then, haptic feedback offers an additional source of information than the static tactile cues related to the object resting on the palm of the non-grasping limb. Thus, the closed-loop and memory-guided grasping conditions described in the preceding paragraph differ not only in terms of the availability of tactile feedback at the time of response cuing but also with regard to the availability of terminal haptic feedback. To our knowledge, the issue of haptic feedback has not been previously explored in a non-visual grasping experiment. Notably, however, work from the visual domain (Schenk, 2012a, 2012b) reported that DF's visual grasping in the absence of haptic feedback was no better than her well-documented visuoperceptual deficits. Accordingly, Schenk proposed that haptic feedback provides a visuo-haptic calibration that can be applied in a predictive fashion to support absolute aperture shaping on future trials (Schenk, 2012a). Although Schenk's findings have received a number of serious challenges (Milner et al., 2012; Whitwell and Buckingham, 2013; Whitwell et al., 2014), his contention represents a salient consideration for tactile-based closed-loop and memory-guided grasping. Thus, we sought to disentangle the putative influence of a memory delay from haptic feedback. To that end, in addition to the memory-guided grasping condition described previously (henceforth referred to as *memory-guided no haptic feedback*: MH-) we included a memory-guided grasping (and manual estimation) condition wherein the experimenter placed the target object between participants' thumb and forefinger once they completed their response to the remembered target location (i.e., *memory-guided with haptic feedback:* MH+). Thus, if haptic feedback engenders a forward calibration based on absolute object information then JNDs in the MH+ grasping condition should be comparable to the closed-loop grasping condition. In other words, the provision of haptic feedback may result in memory-guided grasping responses that violate Weber's law.

[1] In addition to manual estimation and grasping tasks, a classic method of adjustment task was used wherein participants manipulated the size of an object appearing on a computer screen to match the felt size of the target object resting on the palm or forearm of their left limb. The results for the method of adjustment task matched the results for the manual estimation task (Davarpanah Jazi & Heath, 2014).

[2] DF is an extensively studied individual with bilateral lesions to her lateral occipital cortex (James et al., 2003) and a documented visual agnosia. Evidence has shown that DF's lesions impair her visual form perceptions but not her ability to use vision to interact with the environment (for extensive review of this issue see Goodale and Milner (2013)).

2. Method

2.1. Participants

Fifteen individuals (2 males and 13 females: age range = 19–31) years from the University of Western Ontario community volunteered to participate in this study. Participants self-declared that they were right hand dominant with normal or corrected-to-normal vision. Participants signed consent forms approved by the Office of Research Ethics, University of Western Ontario, and this work was completed according to the Declaration of Helsinki.

2.2. Apparatus and procedures

Participants sat for the duration of the experiment in front of a table-top (height = 780 mm, depth = 760 mm, width = 1060 mm) and manually estimated or grasped with their right hand (i.e., thumb and forefinger precision grip) target objects located on the palm of their left hand. Target objects were acrylic blocks painted flat black and were 20, 30 and 40 mm in width and 10 mm in height and depth. Further, target objects were weight-matched (7 g) to preclude size information from being derived from weight cues. In advance of each trial, participants placed the medial surface of their right wrist on a pressure sensitive switch (henceforth referred to as start location) located 200 mm to the right of their midline and 100 mm from the front edge of the table-top. The configuration of the right arm at the start location was such that the shoulder was abducted approximately 40° with elbow and wrist flexed approximately 90° and 45°, respectively. As well, prior to each trial participants were instructed to keep their thumb and forefinger lightly pinched together. In turn, participants positioned their left supinated palm 200 mm to the left of their midline and in the same transverse plane as the right limb's start location. In particular, the left shoulder was at a neutral angle with the elbow flexed at approximately 90°. Computer and auditory events were controlled via MATLAB (7.9.0: The MathWorks, Natick, MA, USA) and the Psychophysics toolbox extensions (ver 3.0) (Brainard, 1997). Moreover, we emphasize that participants wore translucent goggles (PLATO Translucent Technologies, Toronto, ON, Canada) throughout data collection to preclude visual information regarding target objects.

2.3. Manual estimation tasks

For all manual estimation trials a target object was placed on the center of the left palm (denoted via a 10 mm by 10 mm cross) with the target's long-axis oriented in the anteroposterior plane. Subsequently, the experimenter initiated a 4000 ms tactile preview phase after which time participants completed manual estimations in each of four conditions. In the closed-loop with haptic feedback condition (CH+), a high-frequency tone (2900 Hz for 1000 ms) presented after the preview phase signaled participants to manually estimate the width of the target object by separating the distance between the thumb and forefinger of their right hand. Importantly, the response was completed while participants maintained contact with the start location. Once participants indicated that they had appropriately estimated the size of the target object (via verbal prompt), they were instructed to close the separation between their thumb and forefinger and subsequently reach to grasp – but not lift – the target object. At the end of the grasping response participants thumb and forefinger were therefore in contact with the target object and the palm of their non-grasping limb. Participants held the target object for approximately 2000 ms before returning to the start location in preparation for a subsequent trial. For the closed-loop no haptic feedback condition (CH−) participants completed the same experimental procedures as described above with the only exception that a grasping response was not initiated following the manual estimation. Thus, the CH+ but not the CH− condition provided terminal haptic feedback related to object size.

In the memory-guided haptic feedback condition (MH+) the experimenter removed the target object from the left palm following the preview period and initiated a 2000 ms delay. Following the delay, a tone (see above) signaled participants to complete their manual estimation. Once participants were confident that they had accurately estimated the target object size they were instructed to close the separation between their thumb and forefinger and subsequently reach to "grasp" the remembered target object. At the end of the grasping response participants' thumb and forefinger were in contact with the palm of their left hand; however, the absence of the physical target object precluded immediate haptic feedback related to the size of the target object. Thus, to provide terminal haptic feedback the experimenter placed the target object between participants' right thumb and forefinger following movement offset (see definition in Section 2.5). In particular, at movement offset a computer-generated cue signaled the experimenter to place the target object on the lateral surface of participants' left palm and the experimenter subsequently slid the object until it first contacted the thumb of the grasping hand and then positioned the opposite side until it contacted the forefinger of the grasping hand. Participants were then instructed to adjust the target object between their thumb and forefinger to produce a stable grasp (i.e., a grasp that would allow for lifting of the target object). As in the CH+ condition, the target object was then held (but not lifted) for 2000 ms before the participant returned to the start location. In turn, in the memory-guided no haptic feedback condition (MH−) participants remained at the start location following their manual estimation; that is, the response was completed without terminal haptic feedback.

2.4. Grasping tasks

In line with the manual estimation tasks, grasping trials began with the experimenter placing a target object on the palm of participants' left hand after which time a 4000 ms tactile preview was provided. Following the preview, grasping responses were completed in each of three conditions. In the closed-loop haptic feedback condition (CH+) participants reached to grasp – but not lift – the target object in response to the imperative tone. Following contact, participants held the object for approximately 2000 ms before returning to the start location. In the memory-guided haptic feedback (MH+) condition the target object was removed from participants' palm following the preview phase and the imperative tone was provided after a 2000 ms delay. In response to the tone participants completed a grasp to the remembered target location and after achieving their goal location the experimenter positioned the physical target object between participants' right thumb and forefinger as per the description outlined for the MH+ condition in the manual estimation task. For the memory-guided no haptic feedback (MH−) condition participants completed the same procedures as described for the MH+ condition with the only exception being that the experimenter did not position the target object between participants' thumb and forefinger at movement offset. In other words, at the end of the response participants contacted the palm of their non-grasping limb but did not received haptic feedback related to object size. The grasping tasks did not entail the same fully factorial combinations as the manual estimation tasks because it was not possible to structure a closed-loop grasp without terminal haptic feedback (see Fig. 1 for a schematic of the timeline of tactile, auditory and haptic events).

Manual estimation and grasping tasks were performed in

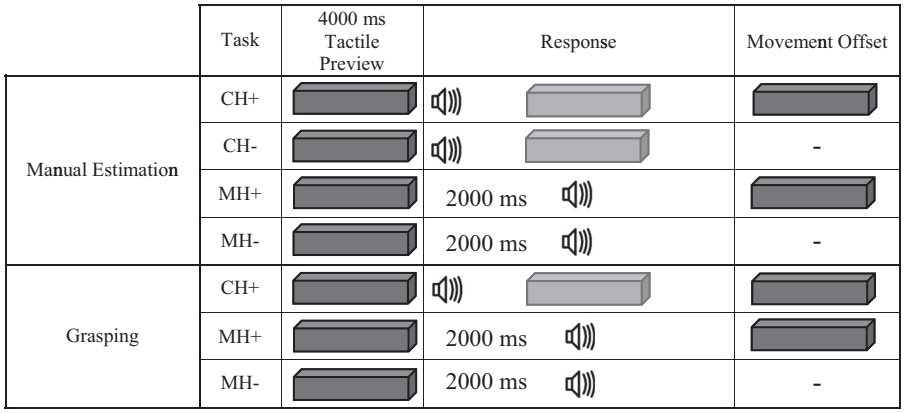

Fig. 1. Schematic representation of the timeline of tactile, auditory and haptic events across manual estimation (CH+ and CH− refer to closed-loop with and without haptic feedback, respectively; MH+ and MH− refer to memory-guided with and without haptic feedback, respectively) and grasping (CH+ refers to closed-loop with haptic feedback; MH+ and MH− refer to memory-guided with and without haptic feedback, respectively) conditions. For all tasks, participants were provided a 4000 ms tactile preview of the target object. For closed-loop manual estimation and grasping tasks, responses were cued immediately following the preview phase and the schematic's transparent depiction of the target object during the 'Response' phase indicates that it remained on the palm of participants' non-grasping limb throughout the response. In the memory-guided tasks, the target object was removed from the palm following the preview phase and a 2000 ms delay was then introduced – removal of the target object precluded real-time tactile feedback at movement cuing and during the 'Response' phase. As well, the depiction of a target object at 'Movement Offset' indicates those conditions wherein terminal haptic feedback was available either immediately after the response (i.e., CH+ tasks) or when provided by the experimenter (i.e., MH+ tasks).

separate sessions conducted at least 24 h apart with tasks being counterbalanced in a sequential order. Within each session, the different manual estimation and grasping conditions were completed in separate and randomly ordered blocks that entailed 15 trials (ordered randomly) to each object size. Thus, 180 and 135 trials were performed for the manual estimation and grasping tasks, respectively. Further, the manual estimation and grasping tasks required approximately 65 and 50 min, respectively to complete. Thus, it was imperative that the different tasks were performed in separate sessions (see above) to reduce participants' physical and mental fatigue.

2.5. Data analysis

The position of the right limb was measured via infrared emitting diodes (IREDs) placed on the lateral surface of the distal phalanx of the forefinger, the medial surface of the distal phalanx of the thumb, and the styloid process of the wrist. IRED position data were sampled at 400 Hz via an OPTOTRAK Certus for 1500 ms following response cuing. IRED position data were filtered offline via a second-order dual-pass Butterworth filter employing a low-pass cutoff frequency of 15 Hz. Subsequently, instantaneous velocities were computed from the position (i.e., displacement) data via five-point central finite difference algorithm. For the manual estimation task, grip aperture (GA) was measured after participants confirmed (via oral response) that they had produced an appropriate size judgment and when offline analysis showed that they had achieved a stable aperture. For the grasping conditions, movement onset was marked when participants released pressure from the start location switch and movement offset was determined when wrist velocity fell below a value of 50 mm/s for 20 consecutive frames (i.e., 50 ms).

2.6. Dependent variables and statistical analyses

For the manual estimation tasks, we examined grip aperture (GA: resultant distance between thumb and index finger) and associated JNDs via 2 (delay condition: closed-loop and memory-guided) by 2 (haptic feedback condition: H+ and H−) by 3 (object size: 20, 30 and 40 mm) repeated measures ANOVA. For the grasping tasks, we examined grasping time (GT: time from movement onset to movement offset), peak grip aperture (PGA: maximum resultant distance between thumb and forefinger) and associated JNDs via 3 (condition: CH+, MH+, MH−) by 3 (object size: 20, 30 and 40 mm) repeated measures ANOVA. Main effects and interactions were considered significant at an alpha level of 0.05 or less. Post-hoc contrasts of within-condition effects of object size were examined via power-polynomials (i.e., trend analysis: see Pedhazur 1997), whereas between-condition effects were decomposed via paired samples *t*-tests.

2.7. Just-noticeable-difference scores (JNDs)

Weber's law states that the smallest detectable difference (i.e., the JND) between an original and a comparator stimulus is proportional to the magnitude of the original stimulus. In the majority of the literature, JNDs are determined via an arbitrary statistical criterion wherein participants reliably discriminate (via oral report or other perceptual judgment) between the original and comparator stimulus (e.g., 75% of trials). Of course, for a grasping task a statistical criterion is not available for JND computation; rather, the JNDs computed here and elsewhere (Davarpanah Jazi and Heath, 2014; Davarpanah Jazi et al., 2014; Ganel et al., 2008a; Ganel et al., 2008b; Heath et al., 2012; Holmes and Heath, 2013; Holmes et al., 2013; Holmes et al., 2011; Pettypiece et al., 2009) represent the within-participants standard deviation of grip aperture. According to Ganel, et al. (2008a) the basis for this JND computation is drawn from the classic method of adjustment wherein variance provides a measure of visuomotor uncertainty "…for which the observer is unable to tell the difference between the size of the comparison and the target object" (p. 600) (see also Marks and Algom, 1998). In demonstration of this approach, Fig. 2 shows manual estimation and grasping task results of an exemplar participant in the CH+ condition. The figure demonstrates that trial-to-trial grip aperture variability (i.e., the JNDs) in the manual estimation task increased linearly as a function of increasing object size, whereas JNDs for the grasping task (computed at PGA) did not vary systematically with object size. Thus, we interpret a linear increase in JNDs with increasing object size as adherence to Weber's law.

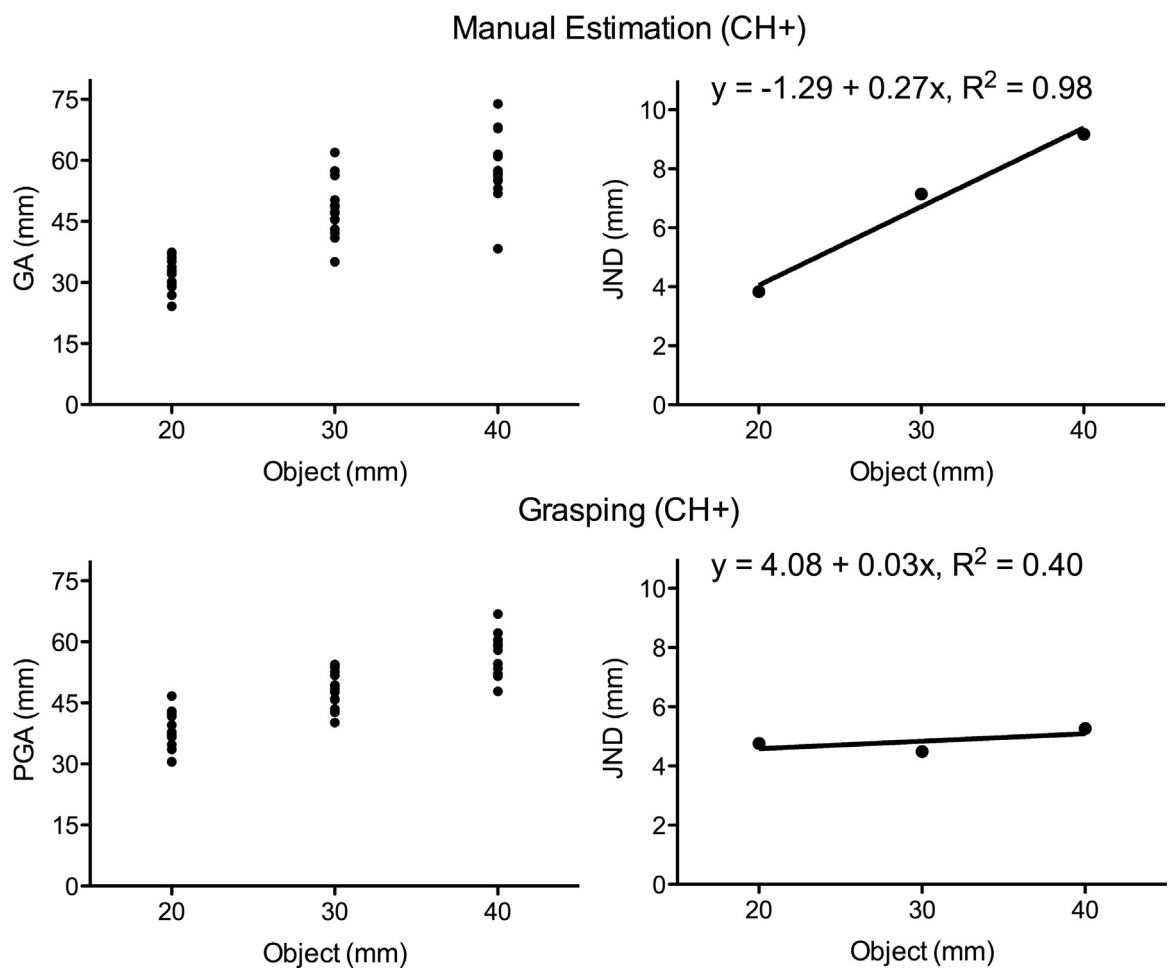

Fig. 2. Exemplar data from a participant in the current study performing CH+ manual estimation and grasping tasks. The upper left panel shows trial-to-trial grip aperture (GA: mm) for manual estimations of differently sized objects and the upper right panel shows the just noticeable difference scores (JNDs) associated with the trial-to-trial GA values. The lower left panel shows trial-to-trial peak grip aperture (PGA: mm) values associated with grasping differently sized objects and the lower right panel shows the computed JNDs associated with the trial-to-trial PGA values. Notably, JNDs for the manual estimation task increased with increasing object size, whereas JNDs for the grasping task did not systematically vary as a function of object size. In other words, CH+ manual estimation and grasping tasks adhered to and violated, respectively, the psychophysical principles of Weber's law.

3. Results

3.1. Manual estimation

Results for GA produced main effects of haptic feedback, $F(1,14)=11.52$, $p<0.01$, and object size, $F(2,28)=149.39$, $p<0.001$. In particular, *H+* trials (39 mm, SD=8) produced larger GAs than their H− trial counterparts (35 mm, SD=9), and Fig. 3 demonstrates that GAs (for all conditions) increased linearly with increasing object size (linear effect: $F(1,14)=178.17$, $p<0.001$). Results for JNDs revealed a main effect of object size, $F(2,28)=48.92$, $p<0.001$, indicating that values (for all conditions) increased linearly with increasing object size (linear effect: $F(1,14)=93.43$, $p<0.001$). Further, we note that JNDs for delay and feedback conditions did not produce main effects ($Fs(1,14)=0.16$ and 1.34 for delay and feedback manipulations, respectively, $ps>0.26$) or higher-order interactions involving object size ($Fs(2,28)=2.44$, 0.43 and 0.35 for delay by object size, feedback by object size and delay by feedback by object size interactions, respectively, $ps>0.10$) (see also Fig. 3). We highlight these null findings because they demonstrate that manual estimations adhered to Weber's law independent of the availability of real-time tactile feedback and terminal haptic feedback.

3.2. Grasping

GT elicited a main effect of condition, $F(2,28)=5.10$, $p<0.05$, such that the CH+ condition (600 ms, SD=25) produced shorter movement durations than MH+ (620 ms, SD=30) and MH− (613 ms, SD=28) conditions ($ts(14)=-2.88$ and -2.26 for CH+ vs. MH+ and CH+ vs. MH-, respectively, $ps<0.05$), and the latter conditions did not reliably differ ($t(14)=1.09$, $p=0.29$). Results for PGA revealed main effects of condition, $F(2,28)=23.24$, $p<0.001$, object size, $F(2,28)=72.62$, $p<0.001$, and their interaction, $F(4,56)=4.46$, $p<0.01$. Fig. 4 shows that PGAs for all conditions increased linearly as a function of increasing object size (linear effect: $Fs(1,14)=64.89$, 86.03 and 59.89 for CH+, MH+ and MH-, respectively, $ps<0.001$); however, PGAs in the CH+ and MH- conditions were respectively larger ($t(14)=4.63$, $p<0.001$) and smaller ($t(14)=3.43$, $p<0.01$) than their MH+ condition counterpart.

Results for JNDs revealed main effects of condition, $F(2,28)=5.63$, $p<0.01$, object size, $F(2,28)=6.71$, $p<0.01$, and their interaction, $F(4,56)=4.82$, $p<0.01$. JNDs for the MH− condition produced an effect for object size, $F(2,28)=10.56$, $p<0.001$), such that values increased linearly with increasing object size (linear effect: $F(1,14)=23.09$, $p<0.001$). In contrast, JNDs for CH+ and MH+ conditions did not reliably differ with object size ($Fs(2,28)=2.05$ and 0.74 for CH+ and MH+, respectively, $ps>0.15$). More directly, JNDs for the CH+ and MH+ conditions did not increase linearly

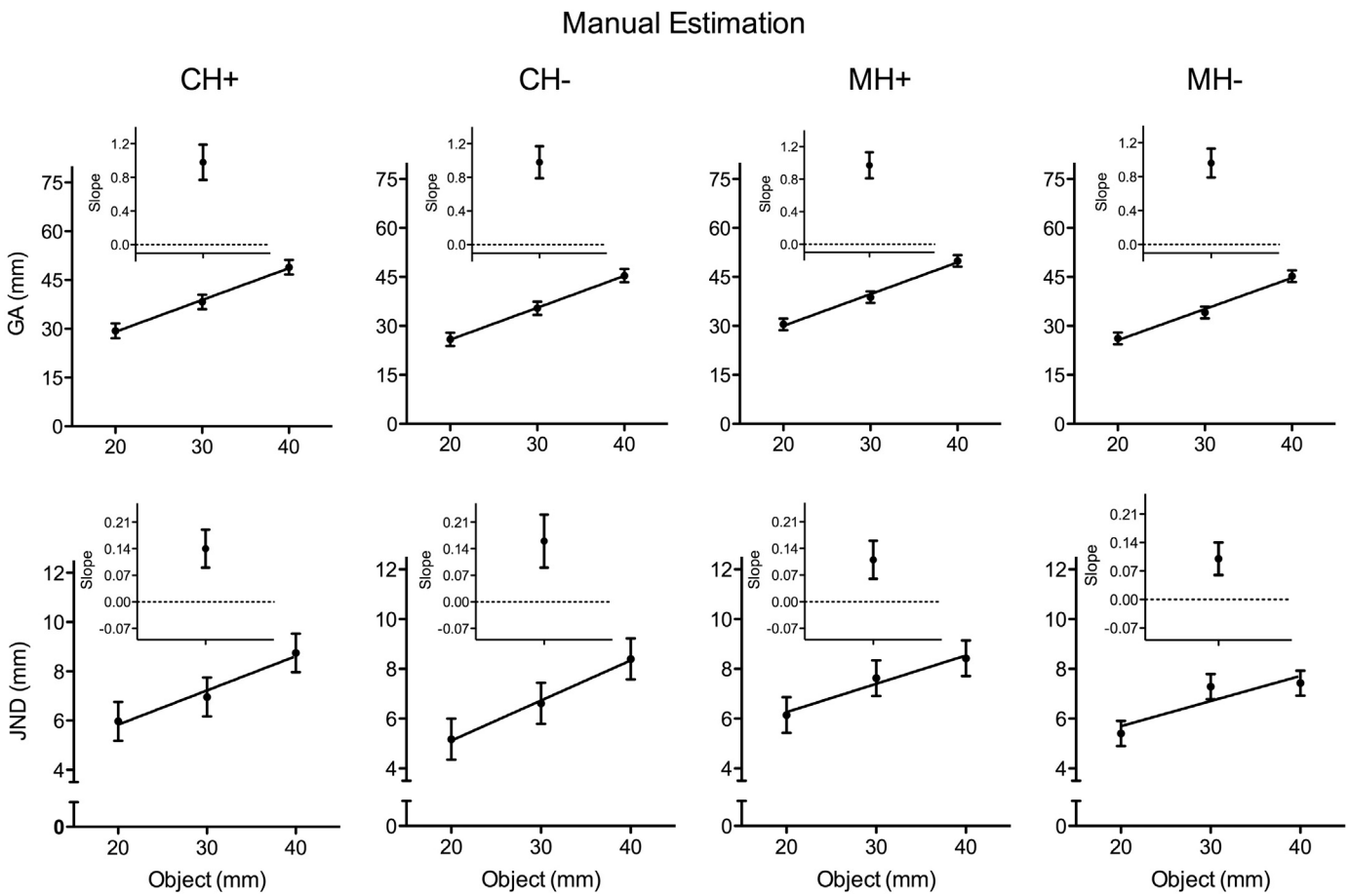

Fig. 3. Manual estimation tasks: mean grip aperture (GA) (upper panels) and just-noticeable-difference (JND) values (lower panels) for the different target objects across closed-loop (i.e., C) and memory-guided (i.e., M) conditions performed with (i.e., H+) and without (i.e., H−) terminal haptic feedback. The solid line in each panel represents the regression line and error bars represent 95% within-participants confidence intervals computed as a function of the mean-squared error term for object size (Loftus and Masson, 1994). The inset figure for each panel represents the mean participant-specific slope representing GA and JND scores to object size and the error bar in each panel represents the 95% between-participant confidence interval. The absence of overlap between the error bar and zero represents a slope value that differs from zero (Cumming and Finch, 2005).

with increasing object size (linear effects: $Fs(1,14)=1.96$ and 1.15, for CH+ and MH+, respectively, $ps>0.18$).

3.3. Participant-specific slopes relating GA/PGA and JNDs to object size

We computed participant-specific slopes relating GA/PGA and JND values to object size. For the manual estimation tasks, slopes were examined via 2 (delay condition: closed-loop and memory-guided) by 2 (haptic feedback condition: H+ and H-) repeated measures ANOVAs, whereas the slopes for the grasping tasks were examined via one-way repeated measures ANOVAs. Manual estimations did not elicit main effects or interactions for GA ($Fs(1,14)=0.06$, 0.03 and 0.04 for main effects of delay and feedback and their interaction, respectively, $ps>0.81$) or JND ($Fs(1,14)=2.74$, 0.04 and 0.56 for main effects of delay and feedback and their interaction, respectively, $ps>0.12$) slopes (see Tables 1 and 2 for GA and JND slopes and associated regression equations). For the grasping tasks, PGA slopes for the CH+ condition (0.55, SD=0.26) were shallower than the MH− (0.68, SD=0.34) and MH+ (0.72, SD=0.30) conditions ($ts(14)=-3.75$ and -2.81 for CH+ vs. MH+ and CH+ vs. MH−, respectively, $ps<0.02$), and the latter tasks did not reliably differ ($t(14)=0.62$, $p=0.54$). In terms of JND slopes, the CH+ (0.02, SD=0.04) and MH+ (0.03, SD=0.11) conditions did not reliably differ ($t(14)=-0.44$, $p=0.67$), and were less than the MH− (0.12, SD=0.09) condition ($ts(14)=-5.38$ and -2.48 for CH+ vs. MH− and MH+ vs. MH−, respectively, $ps<0.03$) (see also Tables 1 and 2).

The inset panels of Figs. 3 and 4 present the mean participant-specific GA/object size, PGA/object size, and JND/object size slopes and their associated 95% between-participant confidence interval for all manual estimation (Fig. 3) and grasping (Fig. 4) tasks. Notably, the absence of overlap between the error bar and zero represents a slope that reliably differs from zero (Cumming and Finch, 2005). As such, the inset panels of Fig. 3 graphically demonstrate that GA/object size and JND/object size slopes for each manual estimation task differed from zero. In turn, the inset panels of Fig. 4 demonstrate that PGA/object size slopes for each grasping task differed from zero and that the JND/object size slope for the MH− condition differed from zero. In turn, JND/object size slopes for the CH+ and MH+ conditions did not reliably differ from zero. Thus, our graphical analyses demonstrate that: (1) manual estimations adhere to Webers law independent of delay and haptic feedback, and (2) grasping responses performed with (CH+ and MH+) and without (MH−) haptic feedback respectively violate and adhere to Weber's law.

4. Discussion

Previous work has shown that tactile-based manual estimations adhere to Weber's law, whereas grasping violates the law (Davarpanah Jazi and Heath, 2014) – a result consistent within the SPM's assertion that tactile perceptions and actions are mediated via relative and absolute cues, respectively (Dijkerman and de Haan, 2007). In the present investigation, we sought to determine whether the

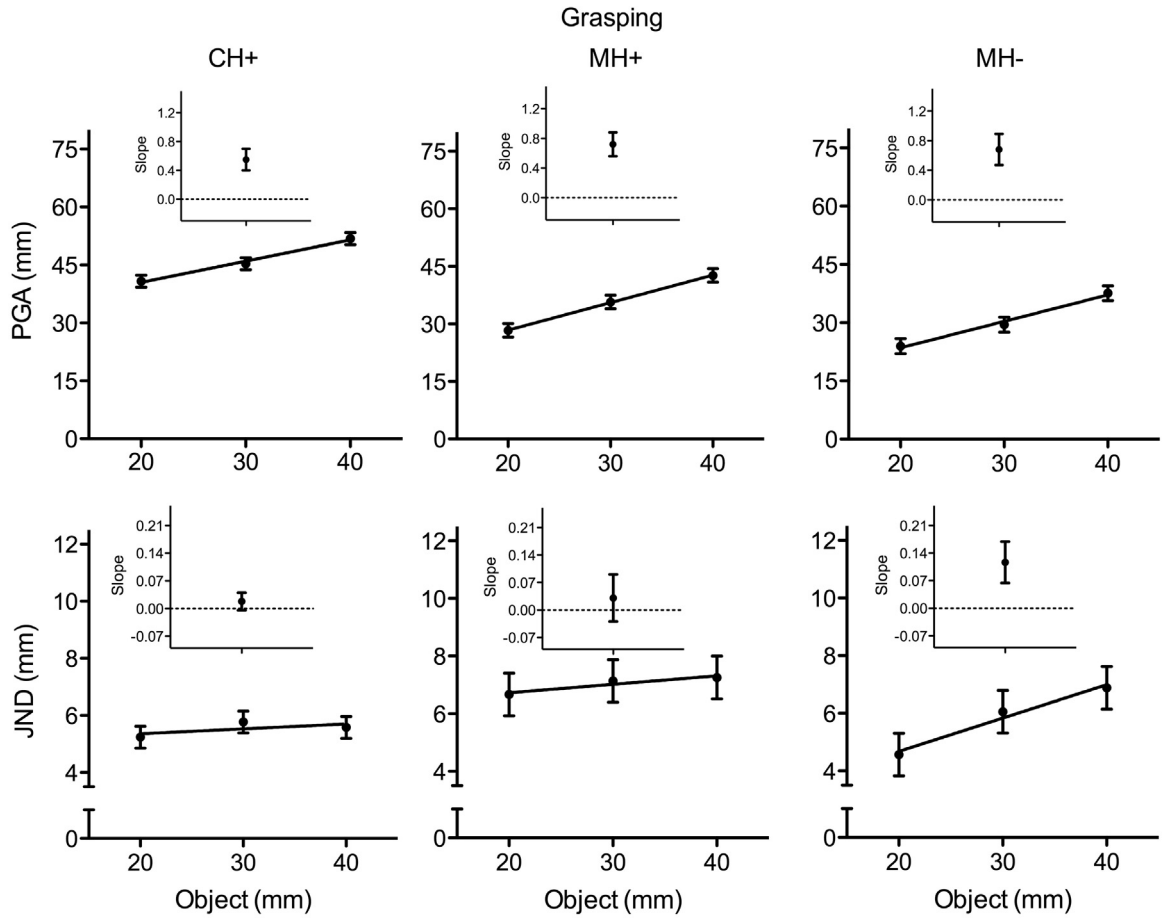

Fig. 4. Grasping tasks: mean peak grip aperture (PGA) (upper panels) and just-noticeable-difference (JND) values (lower panels) for the different target objects in closed-loop (CH+) and memory-guided conditions performed with (MH+) and without (MH−) terminal haptic feedback. The solid line in each panel represents the regression line and error bars represent 95% within-participants confidence intervals computed as a function of the mean-squared error term for object size (Loftus and Masson, 1994). The inset figure for each panel represents the mean participant-specific slope representing PGA and JND scores to object size and the error bar in each panel represents the 95% between-participant confidence interval. The absence of overlap between the error bar and zero represents a slope value that differs from zero (Cumming and Finch, 2005).

Table 1
Linear regression equations and proportion of explained variance (R^2 values) relating grip aperture (i.e., manual estimation tasks) and peak grip aperture (i.e., grasping tasks) to object size for each experimental condition.

	Feedback condition	Delay condition			
		C	R^2	M	R^2
Manual estimation	H+	$y=9.53+0.98x$	(0.99)	$y=10.55+0.97x$	(0.99)
	H−	$y=6.26+0.98x$	(0.99)	$y=6.54+0.95x$	(0.99)
Grasping	H+	$y=29.52+0.55x$	(0.99)	$y=14.04+0.72x$	(0.99)
	H−	–	–	$y=9.90+0.68x$	(0.99)

Note: C=closed-loop; M=memory-guided; H+=haptic feedback; H−=no haptic feedback.

Table 2
Linear regression equations and proportion of explained variance (R^2 values) relating just noticeable difference scores for grip aperture (i.e., manual estimation tasks) and peak grip aperture (i.e., grasping tasks) to object size for each experimental condition.

	Feedback condition	Delay condition			
		C	R^2	M	R^2
Manual estimation	H+	$y=3.06+0.14x$	(0.97)	$y=3.98+0.11x$	(0.97)
	H−	$y=1.89+0.16x$	(0.99)	$y=3.68+0.10x$	(0.80)
Grasping	H+	$y=5.02+0.02x$	(0.41)	$y=6.14+0.03x$	(0.90)
	H−	–	–	$y=2.37+0.12x$	(0.97)

Note: C=closed-loop; M=memory-guided; H+=haptic feedback; H−=no haptic feedback.

introduction of a memory-delay and/or the availability of terminal haptic feedback influence the nature of the information supporting tactile-based manual estimations and grasping. In the below, we first discuss results for our manual estimation tasks prior to outlining the findings for the different grasping conditions.

4.1. Manual estimation: a relative percept supports obligatory judgments of size

GAs for the different manual estimation conditions increased linearly with increasing object size. As such, participants discriminated between the different objects regardless of the memory delay and terminal haptic feedback manipulations used here. It is, however, important to note that the H+ trials produced larger GAs than their H− counterparts. In accounting for this finding, we note that H+ trials involved physically grasping the target object following the manual estimation response. Indeed, grasping a physical object results in the adoption of orthogonal thumb and forefinger approach vectors to avoid an early collision (i.e., safety margin strategy) and to ensure that the forces applied by effectors are parallel to one another at the time of contact (i.e., prevents slipping) (Smeets and Brenner, 1999). Thus, obligatory knowledge that a manual estimation would be followed by a grasping response may have resulted in the adoption of a 'safety margin' task-set similar to that used for grasping. Furthermore, we note that the

proposed safety margin task-set did not influence the representation of object size as the slopes relating GA to object size were consistent across H+ and H− trials.

Results showed that JNDs for the different manual estimation conditions increased linearly with increasing object size; that is, results adhered to Weber's law. Of course, that CH+ and CH− trials adhered to Weber's law supports earlier work by our group (Davarpanah Jazi and Heath, 2014) and provides direct evidence for the SPM's contention that tactile-based perceptions are mediated via relative information. Moreover, the observation that the memory delay in combination with terminal haptic feedback manipulation (i.e., MH+, MH−) also showed adherence to Weber's law – and produced equivalent JND/object size slopes - demonstrates that tactile-based perceptions are supported via an immutable percept. This result is analogous to the properties of the visual system (Goodale and Milner, 2013) and the notion that obligatory judgments are supported via a top-down and experiential-dependent representation of the physical environment.

4.2. Grasping: memory delay and the provision of terminal haptic feedback influences PGA

PGAs across all conditions increased linearly with object size; however, values for the CH+ condition were larger than the MH+ condition, which in turn were larger than the MH− condition. That the memory conditions yielded smaller PGAs is consistent with work from the visual domain showing that responses directed to an area once occupied by, or adjacent to, a target object (so-called pantomime grasping) results in smaller PGAs than grasping a physical target object (Cavina-Pratesi et al., 2011; Davarpanah Jazi et al., 2014; Goodale et al., 1994; Holmes and Heath, 2013; Holmes et al., 2013; Westwood et al., 2000). After all, grasping to an area once occupied by an object obviates the need for orthogonal effector approach vectors because the participant and not the physical properties of the 'remembered' object determine the aperture size necessary for a successful response. In addition, that PGAs for the MH+ condition were intermediary to the CH+ and MH− conditions suggests that the experimenter-induced haptic feedback used here (i.e., the feedback that was provided following the end of the grasping response) allowed for improved-albeit incomplete-calibration of the aperture necessary to grasp a physical target object (see also Bingham et al. 2007). It is, however, less clear why PGA/object size slopes for MH+ and MH− conditions were equivalent and were steeper than the CH+ condition. One account for such a finding is that both MH+ and MH− conditions were associated with an obligatory and perception-based representation of object size. In other words, the PGA/object size slopes can be interpreted to provide *indirect* evidence that a memory delay resulted in the specification of object size via relative cues. Notably, and as will be discussed in detail in the following section, such an explanation is tempered by results from our JND analyses providing *direct* evidence that MH−, but not MH+ trials, adhered to Weber's law. As a final issue in this section, we note that MH+ and MH− conditions produced longer GTs than the CH+ condition. This is an expected finding and is commensurate with work from the visual domain showing that a memory delay elicits longer GTs due to increased uncertainty regarding target location and size (Berthier et al., 1996; Churchill et al., 2000; Holmes et al., 2011; Westwood et al., 2000; Wing et al., 1986). Thus, the analysis of GT evinces that the presence of real-time feedback (tactile or haptic) optimizes grasping efficiency.

4.3. Grasping JNDs: memory delay and the provision of terminal haptic feedback

JNDs for the CH+ condition did not vary systematically with object size and therefore violated Weber's law. This finding supports earlier work by our group (Davarpanah Jazi and Heath, 2014) and supports the SPM's contention that tactile-based actions are mediated via absolute information. In turn, JNDs for the MH− condition increased linearly with increasing object size and therefore adhered to Weber's law. In other words, results evince that memory-guided grasping is supported by the same relative information as manual estimations. To our knowledge such a finding provides the first evidence that the dorsal tactile cortical processing stream operates in the same 'real-time' mode as its visual counterpart (Westwood and Goodale, 2003; for review see Goodale 2011). Interestingly, however, the use of relative cues for memory-guided grasping was modulated by the provision of terminal haptic feedback. In particular, the MH+ condition yielded JNDs that violated Weber's law. More specifically, results show that the MH+ condition yielded JND/object size slopes that were equivalent to the CH+ condition and were shallower than the MH− condition. Thus, results for the MH+ condition provide rule-based evidence that the provision of terminal haptic feedback determines whether relative or absolute information supports memory-guided grasping. As well, we note that the dissociable adherence of MH+ and MH- conditions to Weber's law cannot be accounted for by a speed/accuracy trade-off in motor output variability (Lemay and Proteau, 2001; Meyer et al., 1988). Indeed, if that were the case then the larger PGAs in the MH+ condition would have produced JNDs that were larger than the MH- condition.

4.4. Terminal haptic feedback supports an absolute tactile-haptic calibration

An important issue to address is how terminal haptic feedback supports the absolute specification of object size in a tactile-based memory-guided grasping task. In reconciling this issue we first note that the different target objects used here were randomly varied from trial-to-trial. Thus, participants could not simply rely on haptic feedback from trial N-1 to plan a current grasping response; rather, results suggest a tactile-haptic integration. Second, and as outlined in the Introduction, some work from the visual domain has shown that terminal haptic feedback supports the absolute specification of object size via a visuo-haptic calibration process. In particular, Schenk (2012a) showed that DF's metrical grip aperture scaling was limited to trials wherein she was able to physically grasp a target object; that is, terminal haptic feedback from the target object supported the absolute specification of object size (but see Milner et al., 2012; Whitwell and Buckingham, 2013). In addition, (1) Bingham et al. (2007) showed that responses performed in the absence of online limb vision demonstrated final grip apertures that were comparable to their visually guided counterparts only when terminal haptic feedback was available, and (2) A recent study by our group showed that pantomime-grasping responses performed with and without terminal haptic feedback respectively violated and adhered to Weber's law (Davarpanah Jazi et al., 2014). Accordingly then, some work from the visual domain has proposed that haptic error signals derived from physically grasping a target object support a *visuo-haptic* calibration that is used in a predictive fashion to specify absolute aperture shaping on future trials (Bingham et al., 2007; Davarpanah Jazi et al. 2014; Schenk, 2012a, 2012b).[3] In a similar vein, we propose that the absolute specification of object size in the current experiment was supported via a *tactile-haptic* calibration. More directly, we contend that haptic error signals support the absolute

[3] The calibration between the sensory and motor systems is intrinsic and reflects the normal process of motor skill acquisition (Held and Hein, 1958; see also Redding and Wallace, 2003). Moreover, intrinsic skill acquisition has been shown to occur on a trial-by-trial basis (Laubach et al., 2000).

calibration between tactile cues and motor output. In further support for our proposal, Ernst and Bülthoff's (2004) maximum-likelihood estimation model contends that the effective and efficient execution of actions is associated with a multisensory re-weighting based on the most reliable sensory input. Hence, in our grasping task the static tactile cues associated with the target object resting on the palm of the non-grasping (left) hand may have been less reliable than the absolute haptic feedback associated with physically grasping the target object. Thus, the observation that JNDs for CH+ and MH+ conditions (but not the MH− condition) violated Weber's law provides rule-based evidence of an absolute tactile-haptic calibration specified via the preferential weighting of object size via terminal haptic feedback.

A final issue that we address relates to Whitwell et al.'s (2014) follow-up examination of Schenk's (2012a) work involving DF. In particular, Whitwell et al. showed that DF's ability to scale her grip aperture to object size is independent of whether the visual size of the object matches the haptic feedback derived from 'touching' the object. As such, Whitwell et al. contend that haptic feedback related to the attainment of a movement goal location, and not object size per se, is sufficient to allow DF to calibrate her visuomotor system. Of course, such an explanation cannot be extended to our tactile-haptic study because both the MH+ and MH− conditions provided sensory cues related to the attainment of the movement goal location (i.e., contact of the grasping limb with the palm of the non-grasping limb). Indeed, although both the MH+ and MH− conditions received haptic cues related to the attainment of a movement goal only the ability to physically grasp the target object (i.e., the MH+ condition) resulted in an absolute tactile-haptic calibration. Further, we wish to emphasize that drawing direct corollaries between our work and Schenk (2012a) and Whitwell et al. (2014) must be tempered by the fact that the latter studies are based on: (1) the investigation of a visuo-haptic calibration, and (2) an individual who has experienced a long-term recovery from brain injury. Therefore, our results should be interpreted in the context of how terminal haptic feedback supports the dissociable cortical processing streams underlying somatosensory processing (i.e., the SPM).

5. Conclusions

Manual estimations adhered to Weber's law regardless of the memory and haptic feedback manipulations used here. Thus, manual estimation is a perceptual task mediated via an immutable and relative percept of object size. In addition, memory-guided grasping performed with and without terminal haptic feedback respectively violated and adhered to Weber's law. Such results indicate that terminal haptic feedback supports an absolute calibration between a tactile defined object and the required motor output. What is more, our study highlights that multiple somatosensory cues (i.e., tactile and haptic) support goal-directed grasping.

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